

Interstitial Cystitis / Bladder Pain Syndrome

A detailed plain-language reference — what it is, why it happens, how it's diagnosed, and the full range of treatments — with sources you can check

This report is a companion to the procedure handout. It goes deeper into interstitial cystitis (IC), also called bladder pain syndrome (BPS), so you and your mother have a fuller picture of the condition and the options. It's written to be readable, but it's thorough — feel free to jump to the sections that matter most. Every fact has been checked against authoritative medical sources — the AUA clinical guideline, U.S. government health agencies (NIDDK, FDA, CDC), major medical centers, and peer-reviewed studies — each linked at the end.

How to use this guide

A few quick tips to get the most from it — on a phone, tablet, or computer:

1. **Make the words bigger** if you'd like. On a phone or tablet, place two fingers on the screen and slowly spread them apart. On a computer, hold the **Ctrl** key (or **⌘ Command** on a Mac) and press the **+** key.
2. **The small colored numbers** after sentences each link to the trusted medical source for that fact. Tap or click one to open it, then press your device's **back** arrow to return right to where you were.
3. **The web links in the “Sources” list** at the end work the same way — tap to open, back arrow to come back.
4. **To keep a paper copy**, use the **Print** or **Share** button in your screen's menu and choose Print.
5. **There's no rush.** You don't need to read it all at once. The list of sections near the top (“What's inside”) shows what's in the guide, so you can start with whatever matters most.

What's inside

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1. What IC/BPS is

Interstitial cystitis/bladder pain syndrome is a chronic condition in which the bladder wall is inflamed or irritated, producing pain or pressure in the bladder and pelvic area along with urinary urgency and frequency.^{1,2} By definition the symptoms last more than six weeks and occur *without* a urinary infection or other clear cause — which is a key point: it is not an infection, even though it can feel exactly like one and is often mistaken for a recurring urinary tract infection (UTI). Crucially, it does not respond to antibiotics.^{3,4}

The word "interstitial" refers to the spaces within the bladder tissue, and "cystitis" means bladder inflammation.¹ You'll see the condition written several ways — interstitial cystitis, bladder pain syndrome, painful bladder syndrome, or the combined IC/BPS — which all describe the same disorder.³ It is chronic and tends to be managed over the long term rather than cured, but it is not life-threatening and does not lead to bladder cancer.^{2,5}

✓ Key points at a glance

- A long-term inflammation/irritation of the bladder wall causing pelvic pain, pressure, urgency, and frequent urination.^{1,2}
- Not an infection and not treatable with antibiotics, even though it mimics a UTI.^{3,4}
- Diagnosed by ruling other things out — there is no single definitive test.^{2,5}
- Chronic and managed, not usually cured; symptoms tend to come and go.^{1,5}
- Affects far more women than men, but men get it too and are often underdiagnosed.^{8,21}

2. How common it is

IC/BPS is more common than many people realize. A large U.S. population study (the RAND "RICE" study) estimated that roughly 3.3 to 7.9 million American women — about 2.7% to 6.5% of adult women — have symptoms consistent with the condition.^{24,25} A companion study estimated that on the order of 1 to 4 million American men have symptoms; the male figure is more uncertain and debated, but it suggested that nearly as many men as women may be affected, implying the condition is heavily underdiagnosed in men — where it is often mislabeled as chronic prostatitis.^{26,10} Among people actually diagnosed, women have historically outnumbered men by roughly five to one.^{21,27}

It can begin at any age but most often appears in adulthood, with a median age of onset around 40 (typically between 30 and 70).^{27,19} Anyone — including children — can be affected.^{8,27}

3. What causes it

The exact cause is still unknown, and most experts believe it is not a single disease with one cause but rather a syndrome with several contributing mechanisms that may differ from person to person.^{3,18} The leading theories, often working in combination, include:

- **A defective bladder lining.** The bladder's inner surface is normally coated by a protective layer (the glycosaminoglycan, or GAG, layer). If this barrier is damaged or "leaky," irritating substances in urine can reach and inflame the underlying bladder wall.^{10,18}
- **Mast cell activation.** Mast cells (immune cells that release histamine and other inflammatory chemicals) are often increased and activated in the bladder wall, especially in the ulcer subtype, driving inflammation and pain.^{18,17}
- **Over-sensitized nerves (neurogenic inflammation).** Changes in the bladder's sensory nerves can make normally painless events — like the bladder filling — register as pain.^{3,18}
- **Autoimmune and chronic inflammatory processes,** in which the immune system contributes to ongoing bladder irritation.^{3,18}

Other factors that may raise risk or play a role include a possible genetic predisposition, prior bladder injury or infection, and pelvic floor muscle dysfunction.^{3,2} Importantly, IC is not caused by stress and is not a psychological condition — though stress, like certain foods, can worsen symptoms once the condition is present.⁵

4. The two subtypes

IC/BPS is generally divided into two forms based on what the bladder looks like during cystoscopy. Distinguishing them matters because they behave differently and respond to different treatments.^{17,10}

- **Hunner-lesion (ulcerative) IC.** A minority of patients — commonly cited as roughly 5–10%, though reported rates across studies vary much more widely depending on how carefully clinicians look for them — have Hunner's lesions: distinct reddened, inflamed patches on the bladder wall, with small vessels radiating toward a central scar that can crack and bleed when the bladder is stretched.^{7,27,17} This subtype tends to occur in older patients, comes with more severe symptoms and smaller bladder capacity, is more often linked to autoimmune conditions, and responds better to bladder-directed treatments aimed at the lesions themselves.^{10,17}
- **Non-Hunner (non-ulcerative) IC.** The large majority of patients have this form. The bladder lining looks relatively normal, though tiny pinpoint hemorrhages called glomerulations may appear after the bladder is stretched and drained. Symptoms can be just as severe, but the inflammation seen under the microscope is typically much milder.^{17,10}

① Why the distinction matters

Finding Hunner's lesions changes the plan: these patients are candidates for targeted treatment of the lesions (fulguration/laser or steroid injection) and may improve dramatically afterward. This is exactly the situation the procedure handout describes. Glomerulations alone are non-specific — they can appear in people without IC — so they are not enough to diagnose the condition by themselves.^{13,16}

5. Symptoms and flares

The core symptoms are pelvic or bladder pain (often felt as pressure or tenderness below the belly button), a frequent need to urinate, and a sense of urgency. Pain may worsen as the bladder fills and ease somewhat after emptying. Many people urinate far more often than usual — in severe cases dozens of times a day and night — and pain during or after sex is common.^{2,8} Symptoms range from mild discomfort to severe, life-disrupting pain, and they often fluctuate over time, with periods of worsening ("flares") and periods of relief.^{3,5}

Flares can be set off by specific triggers, which vary widely between individuals but commonly include certain foods and drinks, stress, hormonal changes around menstruation, sexual activity, tight clothing, and seasonal allergies.^{11,12} Identifying personal triggers (see the diet and self-care sections) is one of the most useful things a patient can do.

6. Conditions that often come with it

IC/BPS frequently overlaps with other chronic conditions, particularly other "central sensitivity" or chronic pain syndromes. Recognizing these helps explain why a person may have symptoms beyond the bladder and why a whole-person treatment approach works best.^{12,20} Commonly associated conditions include:

- **Irritable bowel syndrome (IBS)** — reported far more often in IC patients than in the general population (in one study about 39% versus 5%).²⁰
- **Fibromyalgia** and **chronic fatigue syndrome**, both notably more frequent in people with IC.^{20,12}
- **Pelvic floor dysfunction**, where the pelvic muscles are tight or in spasm, which can both mimic and worsen bladder symptoms.^{12,2}
- **Autoimmune conditions** such as lupus, allergies, and (in some patients) endometriosis or vulvodynia.¹²
- **Depression and anxiety.** Living with chronic pain and disrupted sleep takes a real toll, and mood and anxiety symptoms are more common in people with IC. Addressing emotional wellbeing is a legitimate and important part of care, not an afterthought.^{20,12}

7. How it's diagnosed

Because there is no single test that confirms IC/BPS, diagnosis is largely a process of recognizing the characteristic symptom pattern and carefully ruling out other causes (a "diagnosis of exclusion").^{5,10} A typical work-up includes:

- **History and symptom review.** The clinician documents the pattern, location, and severity of pain and the frequency and urgency of urination.¹⁶
- **Bladder (voiding) diary.** Recording fluids taken in, urine passed, and how often, to establish the low-volume, frequent-voiding pattern typical of IC.^{16,6}
- **Pelvic exam** to check for suprapubic and pelvic floor tenderness and to assess the pelvic organs.^{6,16}
- **Urine tests.** Urinalysis and urine culture are essential to rule out infection and check for blood; a culture may be done even when the dipstick is negative, to catch low-level bacteria.^{16,10}
- **Symptom questionnaires** such as the Pelvic Pain and Urgency/Frequency (PUF) scale or the Interstitial Cystitis Symptom Index, used to gauge severity and track response to treatment.¹⁶
- **Cystoscopy and/or urodynamics** — not required to make the diagnosis in straightforward cases, but used when the diagnosis is uncertain or when Hunner's lesions are suspected. Cystoscopy with hydrodistension under anesthesia can reveal glomerulations or Hunner's lesions, and a biopsy may be taken to exclude other conditions including bladder cancer.^{5,16,1}

8. Treatments: the step-by-step ladder

There is no one-size-fits-all cure, and no single treatment works for the majority of patients, so care is individualized and often combines several approaches.^{5,9} The American Urological Association (AUA) organizes treatments into a tiered ladder, generally starting with the least invasive and most reversible options and moving to more aggressive ones only if needed. Patients with Hunner's lesions are an exception — they may move directly to treatment of the lesions.^{5,16,32}

Tier	Main options
First line	Patient education, self-care and behavioral changes (diet, fluid and bladder habits), and stress management — for all patients ⁵
Second line	Pelvic floor physical therapy; oral medications; bladder instillations; pain management ^{5,22}
Third line	Cystoscopy under anesthesia with low-pressure hydrodistension; treatment of Hunner's lesions if found (fulguration/laser and/or triamcinolone injection) ^{5,14}
Fourth line	Bladder Botox (intradetrusor botulinum toxin A); neuromodulation (nerve stimulation) ^{15,5}
Fifth line	Oral cyclosporine A ⁵
Sixth line	Major surgery (urinary diversion, with or without bladder removal) — a rare last resort ^{5,9}

First line — foundations for everyone

Every patient starts here. This includes education about how the bladder works and what is and isn't known about IC, along with self-care and behavioral changes: identifying and avoiding dietary triggers, managing fluid intake, bladder training to gently space out urination, and stress-reduction techniques. These low-risk measures help many people and underpin everything else.^{5,9}

Second line — physical therapy, pills, and instillations

Pelvic floor physical therapy is recommended for patients who have pelvic floor muscle tenderness, and it is one of the better-supported second-line options — a randomized trial found about 59% of patients improved with targeted manual therapy versus 26% with general massage.^{16,5} The right kind is hands-on manual therapy by a trained therapist to release muscle trigger points and connective-tissue restrictions. Importantly, pelvic-floor-*strengthening* exercises (Kegels) are not the goal here; the guidelines specifically advise against them (a strong, Grade-A recommendation), because they can worsen symptoms.^{5,15}

Oral medications may be tried (the AUA lists these alphabetically, with no implied ranking):

Medication	Type / how it may help	Notes
Amitriptyline	Tricyclic antidepressant; calms bladder nerves and blocks pain	Often effective for pain/urgency; can cause drowsiness, dry mouth, weight gain ^{9,22}
Hydroxyzine	Antihistamine; blocks histamine from mast cells	May help, especially with allergies; can cause sedation ^{9,19}
Cimetidine	Histamine (H2) blocker	Some symptom improvement in trials ^{9,22}
Pentosan polysulfate (Elmiron)	Oral "heparinoid" thought to help restore the bladder's protective GAG lining	The only FDA-approved oral drug for IC; taken 100 mg three times daily; may take 3–6 months to work ^{9,6}

⚠ Important: Elmiron and the eyes

Pentosan polysulfate (Elmiron) has been linked to a specific form of retinal damage called pigmentary maculopathy. In 2020 the U.S. FDA updated the drug's label to warn about retinal pigmentary changes.^{28,30} The risk rises with long-term, cumulative use — most reported cases occurred after three or more years, and one study found that roughly one in four people who took it for five years or longer showed significant retinal damage.^{30,29} Possible symptoms include difficulty reading, slow adjustment to dim light, and blurred vision, and the damage has in some cases continued to progress even after stopping the drug.^{29,28} Current guidance is a detailed eye history and a baseline retinal exam (generally within six months of starting), periodic exams thereafter, and reporting any vision changes promptly.^{28,30}

Bladder instillations deliver medication directly into the bladder through a thin catheter, concentrating the drug where it's needed with few whole-body side effects. The main agents are DMSO (dimethyl sulfoxide), heparin, and lidocaine, often combined in a "cocktail" that may also include sodium bicarbonate and a steroid; when DMSO is used, guidelines suggest

limiting how long it dwells in the bladder to roughly 15–20 minutes.^{9,16}

Third line — cystoscopy, hydrodistension, and treating Hunner's lesions

If conservative measures aren't enough, the next step is cystoscopy under anesthesia with short-duration, low-pressure hydrodistension (stretching the bladder with fluid). This can both aid diagnosis and provide temporary symptom relief.^{5,13} If Hunner's lesions are found, guidelines recommend treating them directly — either by destroying them (fulguration with electrocautery or laser) or by injecting the steroid triamcinolone into them, or both. The steroid-injection approach has reported meaningful improvement in roughly two-thirds of treated patients in published series.^{5,14}

This third-line step is precisely the procedure described in detail in the companion handout (cystoscopy + bladder dilation + triamcinolone).

Fourth line — Botox and neuromodulation

Bladder Botox (intradetrusor botulinum toxin A, typically 100 units) is injected into the bladder muscle and can reduce pain and urgency; patients must be willing to accept that temporary self-catheterization may occasionally be needed if the bladder doesn't empty well afterward.^{15,5} **Neuromodulation** uses mild electrical stimulation of the nerves that control the bladder (for example, sacral nerve stimulation). It is usually tested with a trial first, and if the trial helps, a permanent device can be implanted. Neuromodulation tends to help urgency and frequency more than pain.^{5,10}

Fifth and sixth line — reserved options

Cyclosporine A, an oral immune-suppressing medication, is a fifth-line option, considered particularly for Hunner-lesion disease that hasn't responded to lesion treatment. It requires careful monitoring for side effects.⁵ **Major surgery** — such as urinary diversion, sometimes with removal or augmentation of the bladder — is a sixth-line, last-resort option reserved for the small number of people with severe, disabling symptoms who have exhausted everything else.^{5,9}

X Treatments generally not recommended

Guidelines specifically advise against several approaches because they don't help enough or carry undue risk: long-term oral antibiotics, intravesical BCG (outside research settings), high-pressure long-duration hydrodistension, and long-term systemic (oral) steroids. Note this applies to *long-term* steroids — a short course to settle a flare is not ruled out.^{5,16}

9. Diet and nutrition

Diet is one of the most practical levers a patient controls. Research and large patient surveys consistently find that certain foods and drinks trigger flares, although the specific culprits vary from person to person.^{7,23} The most frequently reported triggers are:

- Coffee and tea (both caffeinated *and* decaffeinated), soda, and other carbonated drinks^{7,11}
- Alcohol^{7,23}
- Citrus fruits and juices, and cranberry juice (often recommended for UTIs, but unhelpful here because IC isn't an infection and the acidity can irritate)^{11,7}
- Tomatoes and tomato-based products^{23,7}
- Spicy foods^{7,23}
- Chocolate⁷
- Artificial sweeteners and MSG^{7,23}
- High-acid foods and extra vitamin C²³

✓ A practical approach to eating

- Use an **elimination diet plus a food diary**: remove the common triggers, let symptoms settle, then reintroduce foods one at a time to find personal culprits. Effects can take days to weeks to show, so give each change time.^{11,7}
- The IC diet does **not** have to be severely restrictive — most foods don't bother most people, and many fruits and vegetables are fine.²³
- Staying well hydrated with water generally helps; gentler choices like pear are often better tolerated than acidic juices.⁷
- Over-the-counter products such as calcium glycerophosphate (Prelief) or a little baking soda before a meal may reduce sensitivity to acidic foods for some people.²³

10. Everyday self-care

Alongside diet, several day-to-day strategies form the backbone of first-line management and can meaningfully reduce symptoms:^{5,9}

- **Bladder training** — gently and gradually lengthening the time between bathroom trips, guided by a clinician, to help with frequency.⁵
- **Stress management** — because stress reliably aggravates symptoms, techniques like paced breathing, gentle exercise, mindfulness, and good sleep habits help.^{5,2}
- **Gentle physical activity** and, where relevant, pelvic-floor-aware movement; avoiding things that personally provoke flares (for some, that's strenuous exercise or tight clothing).²
- **Local comfort measures** during flares, such as a warm heating pad over the lower abdomen.¹
- **Smoking cessation**, since smoking can worsen symptoms (and is an independent bladder-cancer risk).²

11. Living with IC: outlook and support

IC/BPS is a chronic condition, and for many people it is a long-term part of life that is managed rather than cured. The encouraging news is that symptoms often fluctuate, most people improve with treatment, and a meaningful share experience spontaneous remissions — in one long-term follow-up study, about half of patients improved over time, with or without ongoing treatment.^{31,1} With a tailored plan, many achieve good symptom control and a full life.^{5,21} Because no single therapy works for everyone, finding the right combination can take patience and some trial and error — that's normal and expected, not a sign of failure.^{5,9}

The condition's impact reaches beyond the bladder. Chronic pain, disrupted sleep, and limits on daily activities can weigh heavily, and depression and anxiety are more common in people with IC. Treating the emotional side — through counseling, support groups, pain specialists, or a multidisciplinary clinic — is a recognized and valuable part of care.^{20,10} Family understanding and support make a real difference. If your mother is finding the emotional burden heavy, it's worth raising with her care team; help is available and effective.

12. Working with the care team

Questions worth bringing to appointments:

- Do I have the Hunner-lesion type or the non-ulcer type, and how does that shape my options?
- Which treatment tier are we starting at, and why?
- If I'm offered Elmiron, what's the plan for monitoring my eyes?
- Can I be referred to a pelvic floor physical therapist?
- How will we measure whether a treatment is working, and how long should I give it?
- What should I do during a flare, and when should I call?
- Is there a multidisciplinary pain or bladder clinic I could be referred to?

struggling emotionally, a doctor or mental-health professional can help, and I'm glad to help you find appropriate support resources if that would be useful.

Sources & further reading

Each claim in this report was checked against the sources below — the American Urological Association guideline, U.S. government health agencies (NIDDK, CDC), major medical centers, the Merck Manual, and peer-reviewed journals. Links are clickable, and the inline numbers throughout the report each link to the matching source.

1. **Cleveland Clinic** — Interstitial Cystitis (Painful Bladder Syndrome). my.clevelandclinic.org/health/diseases/15735-interstitial-cystitis-painful-bladder-syndrome
2. **Johns Hopkins Medicine** — Interstitial Cystitis. hopkinsmedicine.org/health/conditions-and-diseases/interstitial-cystitis
3. **Urology Care Foundation (American Urological Association)** — What is Interstitial Cystitis/Bladder Pain Syndrome? urologyhealth.org/urology-a-z/i/interstitial-cystitis
4. **Harvard Health Publishing** — Interstitial Cystitis. health.harvard.edu/diseases-and-conditions/interstitial-cystitis-a-to-z
5. **American Urological Association** — Diagnosis and Treatment of Interstitial Cystitis/Bladder Pain Syndrome (2022 Guideline). [auanet.org/guidelines-and-quality/guidelines/diagnosis-and-treatment-interstitial-of-cystitis/bladder-pain-syndrome-\(2022\)](https://auanet.org/guidelines-and-quality/guidelines/diagnosis-and-treatment-interstitial-of-cystitis/bladder-pain-syndrome-(2022))
6. **Mayo Clinic** — Interstitial Cystitis: Diagnosis & Treatment. mayoclinic.org/diseases-conditions/interstitial-cystitis/diagnosis-treatment/drc-20354362
7. **National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)** — Eating, Diet & Nutrition for Interstitial Cystitis. niddk.nih.gov/health-information/urologic-diseases/interstitial-cystitis-bladder-pain-syndrome/eating-diet-nutrition
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2. **Interstitial Cystitis Association** — Associated Conditions. ichelp.org/understanding-ic/learn-about-ic/associated-conditions
3. **Interstitial Cystitis Association** — Cystoscopy with Hydrodistention. ichelp.org/cystoscopy-with-hydrodistention
4. **Urology Times** — Steroid injection efficacious for Hunner's lesions (Washington University series). urologytimes.com/view/steroid-injection-efficacious-for-hunners-lesions
5. **Urology Times** — AUA IC/BPS Guideline Update Highlights New Data. urologytimes.com/view/aua-icbbs-guideline-update-highlights-new-data
6. **Journal of Urology (AUA)** — Diagnosis and Treatment of IC/BPS (guideline article). auajournals.org/doi/10.1097/JU.0000000000002756
7. **Whitmore et al., International Journal of Urology (2019)** — Hunner lesion versus non-Hunner lesion IC/BPS. onlinelibrary.wiley.com/doi/10.1111/iju.13971
8. **Diagnostics, MDPI (2022)** — Urinary Biomarkers in IC/BPS and Therapeutic Outcome (pathophysiology & subtypes review). mdpi.com/2075-4418/12/1/75
9. **Medscape / eMedicine** — Interstitial Cystitis: Practice Essentials, Workup, and Medication. emedicine.medscape.com/article/2055505-overview
0. **Nickel et al., Journal of Urology (2010)** — IC/PBS and Associated Conditions (IBS, fibromyalgia, chronic fatigue). pubmed.ncbi.nlm.nih.gov/20719340
1. **Interstitial Cystitis Network** — IC Prevalence & Epidemiology (summarizing RAND and managed-care studies). icnetwork.org/interstitial-cystitis-prevalence-epidemiology
2. **U.S. Pharmacist** — New Approaches in Managing IC/BPS (oral agents & instillations). uspharmacist.com/article/new-approaches-in-managing-interstitial-cystitis-bladder-pain-syndrome
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4. **NIDDK Central Repository** — RAND Interstitial Cystitis Epidemiology Study (RICE): women's prevalence (2.7–6.5%; 3.3–7.9 million). repository.nidk.nih.gov/study/52
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2. **American Urological Association** — Diagnosis and Treatment of IC/BPS (2022), full guideline PDF. auanet.org/documents/Guidelines/PDF/ICBPS_Guideline.pdf